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POLYMERIC NANOPARTICLES FOR DIABETIC WOUND HEALING: CURRENT ADVANCES AND TRANSLATIONAL PROSPECTS

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ABSTRACT:

Diabetic wounds constitute a significant consequence of diabetes, characterized by chronic inflammation, compromised angiogenesis, microbial colonization, increased oxidative stress, and dysregulated extracellular matrix remodelling. Conventional therapeutic approaches show limited success, highlighting an unmet need for more efficient and targeted interventions. Polymeric nanoparticles (PNPs) have recently emerged as a viable nano-biomaterial platform for the delivery of medicines, biomolecules, and bioactive agents, offering increased stability, controlled release, greater bioavailability, and superior retention at wound sites. Natural and synthetic polymer-based nanoparticles, such as chitosan, hyaluronic acid, PLGA, PEG, and PCL, can be engineered to exert multi-functional actions, including antibacterial activity, antioxidant effects, modulation of inflammatory cytokines, and stimulation of angiogenesis. This review highlights current advances in polymeric nanoparticle-based strategies for diabetic wound healing, summarizing their design principle, therapeutic mechanism and clinically relevant outcomes. Furthermore, the translational prospects, regulatory considerations, and future challenges in developing a PNP-based smart wound care system will be discussed. Overall, polymeric nanomedicine represents a promising translational approach for overcoming biological barriers in diabetic wound repair and accelerating tissue regeneration.

Keywords: *Polymeric nanoparticle, Diabetic wound healing, Chronic wound, Angiogenesis, Targeted drug delivery.*

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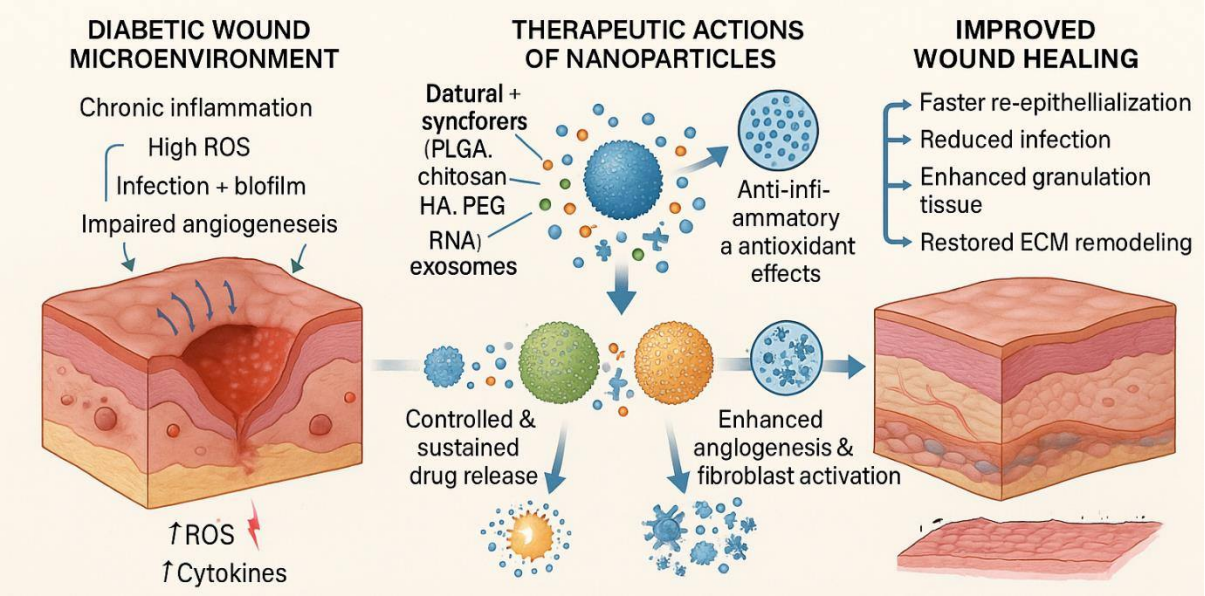


Figure 1 Graphical Abstract

1. Introduction

Diabetes mellitus (DM) is a metabolic disorder marked by disrupted glucose metabolism. There are two forms of diabetes mellitus: type I, resulting from autoimmune destruction of pancreatic beta cells, and type II, characterized by insulin resistance and subsequent pancreatic beta cell dysfunction (Cahill et al., 2019). Elevated blood glucose levels, defined by insulin resistance or diminished insulin secretion, lead to many complications, including diabetic nephropathy, neuropathy, cardiomyopathy, hearing impairment, and cutaneous ulcers. The global prevalence of diabetes is increasing, impacting hundreds of millions of individuals due to lifestyle modifications and cultural advancements (Okonkwo & DiPietro, 2017)(Kesharwani et al., 2018). Diabetic wounds pose a significant concern for individuals with diabetes and have become a critical threat to global health. Chronic, intractable ulcers may arise from persistent infections caused by cellular dysfunction, microcirculatory abnormalities, elevated oxidative stress, and hypoxia, which are the primary contributors to amputation and disability in diabetic wounds (S.-J. Li et al., 2016). Standard clinical treatments encompass wound management, skin grafting, surgical debridement, glycemic control, and hyperbaric oxygen therapy (Shen et al., 2016)(L. Chen et al., 2017)(Dixon & Edmonds, 2021). The therapeutic effectiveness of these clinical therapies on diabetic wound healing is limited, although they may enhance symptom management.

Furthermore, due to the chronic nature of diabetic therapy, patients are increasingly susceptible to secondary complications, which significantly detriment both physical and mental health. Consequently, there is an urgent

demand for contemporary treatments for diabetic wounds that are potentially successful, non-invasive, and devoid of scarring. Published research indicates that nanoparticles have traditionally served as physiologically active delivery methods for wound healing (Blanco-Fernandez et al., 2021). The innovative potential of nanoparticles developed through nanotechnology to elucidate the biological environment and to offer tailored therapeutic strategies for wound healing has garnered interest (Haque et al., 2021). Besides being diminutive, uniform, and readily assimilated by cells, they also exhibit exceptional biocompatibility and targeting capabilities, proficiently regulating the delivery and release of pharmaceuticals (Qi et al., 2021). The properties of nanoparticles (NPs) serve as the main driving force behind the development of novel nanotechnology-based platforms that markedly affect angiogenesis, collagen production, and extracellular matrix (ECM) formation, all of which are essential for promoting wound healing. Nanoparticles (NPs) and diverse NP-based platforms might affect the wound healing process by influencing many cellular and molecular pathways, hence promoting diabetic wound repair due to their intrinsic nanoscale characteristics. Because of their potential to improve the stability and bioavailability of many bioactive substances, they offer a promising strategy for the treatment of diabetic wound healing (Stark et al., 2015).

2. Diabetic Wound-Healing Process

The restoration of the physiological process in wound healing is intricate and dynamic, comprising four stages: hemostasis, inflammation, proliferation, and remodeling. Vascular constriction and thrombin activation augment platelet aggregation, leading to thrombus formation and the release of certain pro-inflammatory mediators, such as cytokines, initiating the hemostasis phase (0 hours post-injury). Mast cells release 5-hydroxytryptamine and histamine, enhancing vascular permeability at the wound site and facilitating the recruitment of neutrophils, monocytes, and chemokines to the lesion, so initiating an inflammatory response that lasts for one to three days (Mirza & Koh, 2015). In the proliferative phase (4–21 days), lymphocytes and macrophages infiltrate the wound to combat infection and degrade cellular debris, hence releasing growth factors and particular cytokines that promote cellular division (Han & Ceilley, 2017). During the remodeling phase, which persists for a minimum of 21 days, new epithelial tissue develops at the injury site. Moreover, fibroblasts transform into muscle fibroblasts in substantial numbers, new blood vessels are produced, and the wound contracts. A scar develops because of this remodeling (Choudhury et al., 2020)(Y. Wang et al., 2021)(Bai et al., 2020).

The damaged organism is expected to undergo standard wound processing under normal physiological conditions. Nevertheless, a chronic wound (such as a diabetic wound) forms when aberrant elements

continuously encourage or impede recovery. Diabetic wounds can become chronic and challenging to heal for several reasons.

First, diabetic wound damage is mostly caused by hypoxia. Because oxygen induces angiogenesis, enhances immunological function, and promotes fibroblast proliferation, its significance for wound healing is increasingly apparent (Choudhury et al., 2020). However, people with diabetes utilize more oxygen at wound sites, in addition to having an inadequate oxygen supply (Desmet et al., 2018). Hypoxia increases levels of oxygen-free radicals, which prolong effective wound healing by intensifying the inflammatory response.

Secondly, disrupted pathological mechanisms, persistent infections, and dysregulated angiogenic factors including transforming growth factor-α (TGF-α), TGF-β, fibroblast growth factor-2 (FGF-2), vascular endothelial growth factor (VEGF), epidermal growth factor (EGF), hypoxia-inducible factor (HIF-1α), and vascular inhibitory factors (such as platelet-activating protein, endothelial inhibitor, and vasopressor)—are marked by reduced local angiogenesis and an inadequate blood supply in diabetic wounds. Thus, a correlation may exist between protracted wound healing and persistent wound inflammation (Choudhury et al., 2020)(Bai et al., 2020). Additionally, diabetics experience decreased blood flow due to macroangiopathy. Microvascular dysfunction arises from diminished nitric oxide (NO) production and altered diastolic vascular function, driven by impaired capillary regulation and reduced nitric oxide synthase (NOS) levels.

Third, diabetes wounds promptly transition into a chronic inflammatory phase, distinct from the typical healing process. Chronic inflammation, marked by an excessive accumulation of M1 macrophages, markedly diminishes fibroblast differentiation into myofibroblasts, inhibits TGF-β type II receptor expression, and reduces collagen synthesis, potentially arising from uncontrolled inflammation and the persistence of inflammatory cells at the wound site. This will impede the skin's remodeling phase (Bai et al., 2020).

Fourth, keratinocyte migration is regulated by matrix metalloproteinases (MMPs), which degrade collagen. Elevated glucose levels lead to the overexpression of MMPs, which impedes the healing of diabetic wounds by exacerbating the local inflammatory response and destroying the extracellular matrix, growth factors, growth factor receptors, integrins, and their receptors (N. Li et al., 2020).

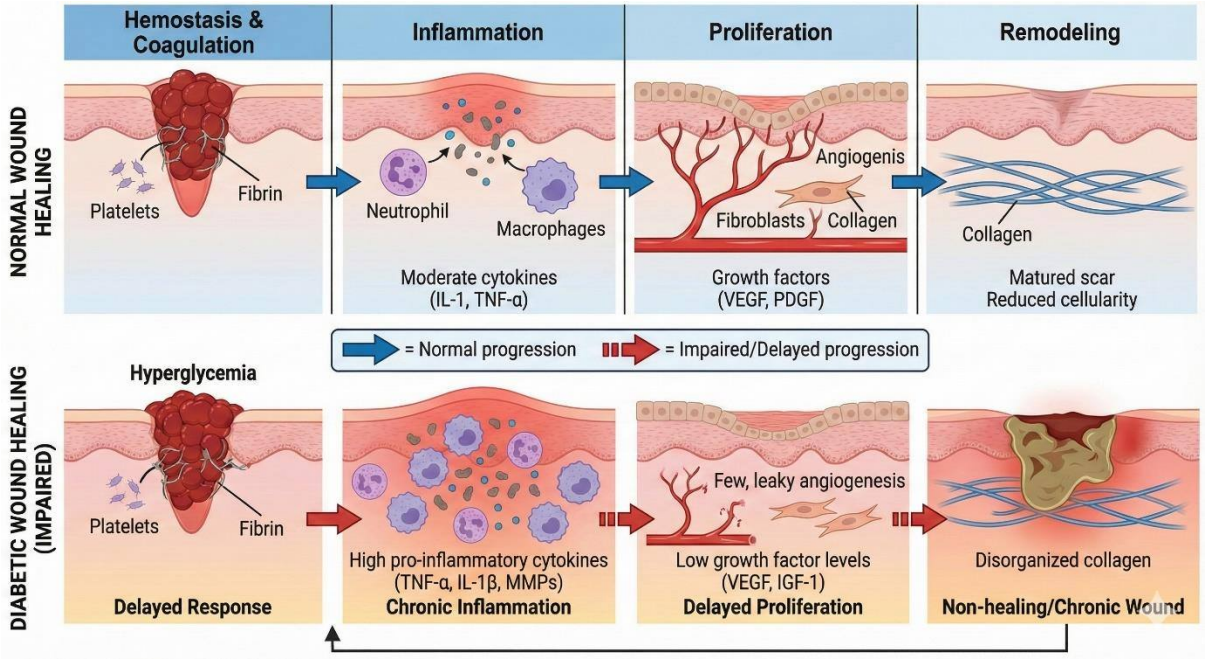


Figure 2 Schematic of the Diabetic Wound Healing Cascade

3. Mechanisms of Polymeric Nanoparticles in Wound Healing

In Figure 2, Polymeric nanoparticles (PNPs) mitigate the limitations of conventional therapies, including insufficient localized action, rapid degradation, and suboptimal drug penetration, thereby facilitating expedited healing of diabetic wounds (Qin et al., 2022). Their nanoscale dimensions, extensive surface area, diverse surface chemistry, and ability to encapsulate many therapeutic agents facilitate targeted and sustained delivery, modulation of the wound microenvironment, and enhanced tissue regeneration (Augustine, Kalarikkal, et al., 2014)(Patra et al., 2018). This outlines the primary methods by which polymeric nanoparticles facilitate wound healing (Gurtner et al., 2008).

3.1 Controlled and Long-lasting Drug Release

The ability of polymeric nanoparticles to regulate and long-term retain medications in the body is one of their strongest features for treating diabetic wounds (J. S. Boateng et al., 2008). The rapid degradation of therapeutic drugs in chronic wounds by proteases, oxidative stress, and microbial enzymes makes it challenging to maintain adequate drug concentrations at the wound bed (Moura et al., 2013)(R. Zhao et al., 2016). Polymeric nanoparticles allow medications such as growth hormones, antibiotics, anti-inflammatory medicines, or antioxidants to be released gradually and steadily over extended periods, preventing premature breakdown (Veith et al., 2019). As a result, there is less need for repeated administrations, the therapeutic dose remains

constant, and the chance of systemic side effects is reduced. The polymer's molecular weight, hydrophobicity, crosslinking density, or breakdown rate can all be altered to adjust release kinetics (Kalashnikova et al., 2015). Some of the best biodegradable polymers are PLGA, chitosan, PCL, and PEG, as they allow for controlled erosion and diffusion (Jayakumar et al., 2011). Drug-releasing polymeric nanoparticles help patients continue taking their prescriptions, promote wound healing, and ensure they continue to receive the necessary care.

3.2 Anti-Inflammatory Modulation

In diabetic wounds, persistent and dysregulated inflammation substantially hinders efficient healing (Singh et al., 2017). Excessive levels of pro-inflammatory cytokines, such as TNF-α, IL-1β, and IL-6, impede fibroblast activity, prolong tissue regeneration, and maintain immune activity. To alter this inflammatory microenvironment, polymeric nanoparticles are crucial because they enable the direct delivery of siRNA, cytokine inhibitors, natural bioactives, and anti-inflammatory medications. In addition to promoting macrophages to transition from the M1 (pro-inflammatory) to the M2 (pro-healing) phenotype, these nanoparticles can reduce the activation of pro-inflammatory pathways. For wound healing to proceed to the proliferative phase, this is crucial. The ability of surface-engineered nanoparticles to facilitate uptake by immune cells can help control the immune system in a specific region without weakening it overall. By restoring the balance of inflammation, polymeric nanoparticles create favorable conditions for angiogenesis, ECM remodeling, and overall wound closure.

3.3 Antioxidant Activity

Diabetic wounds are particularly vulnerable to oxidative stress. Overproduction of reactive oxygen species (ROS) can disrupt cellular processes, damage extracellular matrix components, inhibit angiogenesis, and delay re-epithelialization (Schäfer & Werner, 2008)(Guo & Dipietro, 2010). Polymeric nanoparticles offer a powerful approach to address this imbalance by delivering antioxidants to the wound site in a controlled, sustained manner. Curcumin, resveratrol, quercetin, and other polyphenols are examples of bioactives that can be safely encapsulated in nanoparticles, which prevents them from degrading too quickly and increases their bioavailability to the body (Hewlings & Kalman, 2017)(J. Dong et al., 2023). Additionally, thioketal, polythioketal, or phenylboronic ester-based carriers are examples of ROS-responsive polymers that enable smart release (S. Wang et al., 2024). This implies that the nanoparticles degradeonly in settings with elevated ROS, releasing antioxidants precisely when and where they are required. This focused ROS scavenging accelerates the transition from the inflammatory to the proliferative phase, reducing cell damage, promoting

fibroblast growth, and aiding collagen formation (Birben et al., 2012). Redox equilibrium is restored by polymeric nanoparticles, which significantly enhance the quality and rate of diabetic wound healing (Ezhilarasu et al., 2020).

3.4 Angiogenesis Promotion

Diabetic wounds are characterized by impaired angiogenesis, leading to delayed granulation tissue formation, poor nutrient supply, and insufficient oxygenation. By delivering pro-angiogenic bioactives in a targeted and controlled manner, polymeric nanoparticles provide an efficient platform to restore angiogenic balance (Z. Chen et al., 2025). Growth factors that are highly unstable when unbound include PDGF, VEGF, FGF, and angiopoietin (J. Song et al., 2025). They become more stable, remain active for longer, and do not degrade rapidly at the wound site when enclosed in polymeric nanoparticles. Furthermore, gene vectors or microRNAs (e.g., miR-126, miR-21) that promote endothelial cell migration, proliferation, and tube formation can be delivered by nanoparticles (Lyttle et al., 2023). Certain polymers, such as chitosan and variants of hyaluronic acid, naturally promote angiogenesis by altering the matrix's structure and facilitating endothelial cell adhesion (Elshazly et al., 2025)(Mahmoud et al., 2024). To ensure accurate and prompt therapeutic action, stimuli-responsive nanoparticles can also release angiogenic drugs in response to local cues like pH, ROS levels, or enzymatic activity. Polymeric nanoparticles greatly increase oxygen delivery, accelerate granulation tissue formation, and promote rapid re-epithelialization in diabetic wound healing by stimulating neovascularization.

3.5 Antimicrobial actions

The healing process in diabetic wounds is severely hampered by microbial infection and persistent biofilms. Microbes can easily colonize the wound microenvironment due to its high glucose levels and inadequate immune response (Z. Chen et al., 2025)(Qin et al., 2022). The complex, protective communities of bacteria known as biofilms hinder the effectiveness of conventional antibiotics (Jing et al., 2024). By enabling the efficient penetration and controlled distribution of antimicrobial drugs, polymeric nanoparticles offer a flexible solution to these problems (Zheng et al., 2025). To increase stability and reduce the likelihood of resistance, antimicrobials, metal ions (Ag^{+}, Zn^{2+}, Cu^{2+}), antimicrobial peptides (AMPs), and plant-derived antimicrobials can all be encapsulated within nanoparticles (Ezhilarasu et al., 2020)(Zheng et al., 2025). Because cationic polymers like chitosan degrade bacterial cell membranes, increase permeability, and prevent biofilm formation, they naturally exhibit antibacterial properties (Qin et al., 2022). Additionally, nanoparticles with surface modifications or ligand decorations enhance targeting and penetration into pre-existing biofilms,

facilitating the disintegration of the extracellular polymeric substance (EPS) matrix (Obuobi et al., 2022). When antimicrobials are required, certain polymeric carriers are designed to respond to specific indicators of illness, such as low pH or elevated bacterial enzyme levels. Together, these mechanisms reduce the number of microorganisms, prevent biofilm formation, and clean the wound bed, all of which promote tissue regeneration (Qin et al., 2022)(Zheng et al., 2025).

3.6 Promotion of Fibroblast Migration and ECM Remodeling

The proliferative phase of wound healing necessitates modifications to the extracellular matrix (ECM) and the migration of fibroblasts. This is often fairly difficult in diabetic wounds (Z. Chen et al., 2025)(Burgess et al., 2021). These processes are significantly enhanced by polymeric nanoparticles, which introduce bioactive chemicals that promote collagen synthesis, fibroblast proliferation, and matrix organization. Growth factors such as TGF-β, EGF, and PDGF can be protected from rapid degradation and maintained at the wound bed by encapsulation, which aids fibroblast recruitment and activation (Ezhilarasu et al., 2020). To restore balance between the degradation and synthesis of extracellular matrix (ECM), nanoparticles can modify the production of matrix metalloproteinases (MMPs) and their tissue inhibitors (TIMPs). In chronic diabetes wounds, this balance is often disrupted (Burgess et al., 2021). Certain natural polymer-based nanoparticles, such as chitosan, gelatin, or hyaluronic acid, naturally aid cell adhesion, motility, and matrix deposition by interacting with fibroblasts, as they do in the body (Kumar, Mahmood, et al., 2024). Polymeric nanoparticles facilitate expedited and improved tissue healing by enhancing collagen deposition, promoting granulation tissue formation, and fostering more organized extracellular matrix remodeling (Jayakumar et al., 2011).

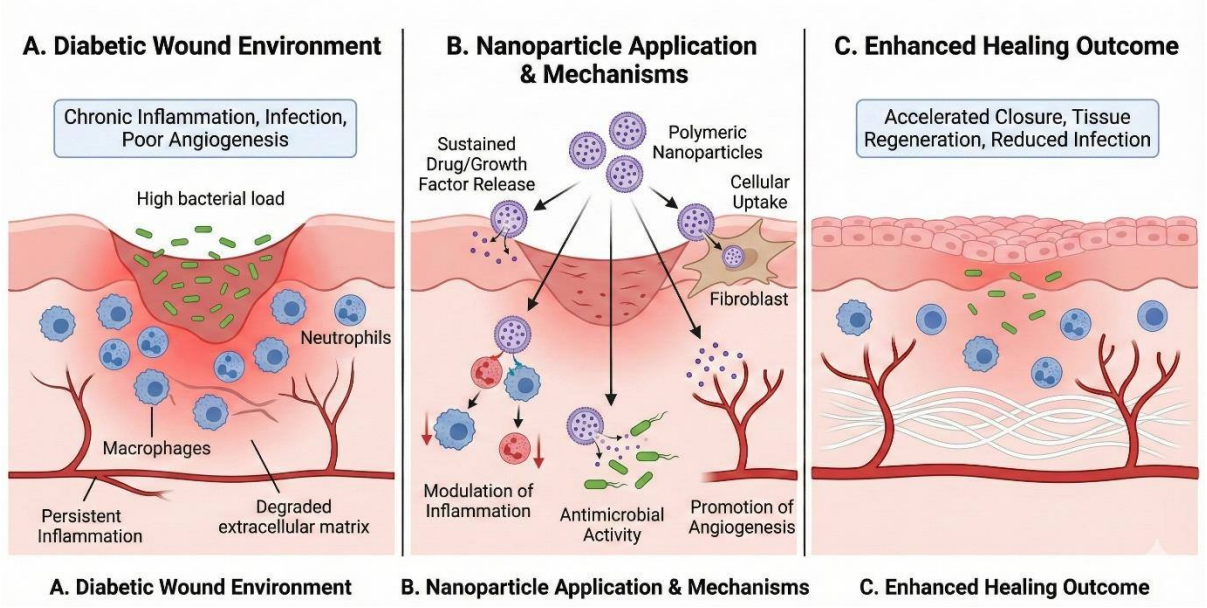


Figure 3: Polymeric nanoparticles speed up the healing of diabetic wounds by using several synergistic methods

4. Polymeric Nanoparticle-Based Therapeutics in Diabetic Wound Healing

Polymeric nanoparticles (PNPs) have emerged as highly effective treatments for diabetic wounds due to their ability to encapsulate various bioactive compounds, prevent their degradation, and deliver them in a targeted and controlled manner (Qin et al., 2022). Figure 3 demonstrates several synergistic effects of these nanoparticles, including regulation of inflammation, defense against oxidative stress, promotion of bacterial death, support for angiogenesis, and support for ECM remodeling. Their ability to treat the complex pathophysiology of diabetic wounds makes them ideal. Depending on their therapeutic payload and functional results, polymeric nanoparticle-based therapies can be broadly categorized in several ways (Z. Chen et al., 2025)(Karri et al., 2016)(Gainza et al., 2014).

4.1 Growth Factor-Loaded Polymeric Nanoparticles

Growth factors (GFs) are crucial for tissue healing, but because they degrade quickly, have a limited half-life, and poorly penetrate chronic wound tissue, they are not very effective in treating diabetic wounds (Barbul, 2008)(Brem & Tomic-Canic, 2007). GF stability, bioavailability, and localized retention at the wound site are significantly enhanced by the protective and long-lasting platform that polymeric nanoparticles provide (Qin

et al., 2022). To maintain biological activity during the healing process, these nanocarriers incorporate several bioactive growth factors and release them in a regulated manner (J. Boateng & Catanzano, 2015). Common growth factors include Vascular Endothelial Growth Factor (VEGF), which facilitates angiogenesis and endothelial cell migration; Platelet-Derived Growth Factor (PDGF), which stimulates fibroblast proliferation and extracellular matrix (ECM) synthesis; Epidermal Growth Factor (EGF), essential for rapid keratinocyte proliferation and re-epithelialization; Transforming Growth Factor-β (TGF-β), which encourages granulation tissue formation and collagen maturation; and Basic Fibroblast Growth Factor (bFGF), which promotes angiogenesis and fibroblast activation (Barrientos et al., 2008). By ensuring prolonged and targeted delivery of these delicate biomolecules, polymeric nanoparticles overcome the biochemical limitations of diabetic wounds and accelerate tissue regeneration (Y. Song et al., 2021).

4.2 Antimicrobial and Antibiofilm Nanoparticles

Polymeric nanoparticles offer a powerful means of overcoming the chronicity of diabetic wounds, which is largely caused by microbial infection and biofilm formation (Burgess et al., 2021)(Qin et al., 2022). Antimicrobial agents, such as metal ions, antibiotics, antimicrobial peptides, and phytochemicals, are more stable, easier to deliver to the wound site, and more effective there when encapsulated in polymeric carriers. Bacterial cell walls are spontaneously broken down by cationic polymers such as chitosan (Raafat & Sahl, 2009). However, the extracellular polymeric substance (EPS) of the biofilm interacts more easily with surface-functionalized nanoparticles, allowing them to penetrate deeper into the biofilm and kill more bacteria (Le et al., 2021). Smart polymeric nanoparticles can release antimicrobials in response to infection-specific cues, such as bacterial enzyme activity or an acidic pH. Systemic toxicity and resistance are less likely as a result (Zheng et al., 2025). By efficiently dissolving biofilm structure, inhibiting microbial development, and reducing the bioburden, antimicrobial and antibiofilm polymeric nanoparticles expedite wound bed preparation and healing in diabetic wounds (Jing et al., 2024)(Burgess et al., 2021).

4.3 Antioxidant and Anti-inflammatory Nanoparticles

Chronic inflammation and oxidative stress are two interrelated clinical conditions that substantially impede the healing of diabetic wounds (Schäfer & Werner, 2008)(Burgess et al., 2021). Increased reactive oxygen species (ROS) cause cellular damage, impair fibroblast function, and amplify inflammatory signals. The lesion's healing is hindered by the persistent release of pro-inflammatory cytokines, including TNF-α, IL-1β, and IL-6 (Guo & Dipietro, 2010). By simultaneously addressing oxidative stress and inflammation through

controlled, targeted injections of antioxidant and anti-inflammatory chemicals, polymeric nanoparticles provide a dual therapeutic strategy (Z. Chen et al., 2025).

The stability, solubility, and bioavailability of natural antioxidants such as curcumin, resveratrol, quercetin, and other polyphenols are significantly enhanced by nanoparticle encapsulation. Additionally, it permits prolonged release at the wound site (L. Wang et al., 2024).

Polymers that respond to stimuli are far more sophisticated. For instance, ROS-responsive carriers, such as thioketal- or phenylboronic-ester-based nanoparticles, release their therapeutic load at the exact moment when oxidative stress peaks, since they degrade only in high-ROS settings. Treatment precision is increased, and off-target effects are decreased thanks to this tailored responsiveness (Liu et al., 2023). The modest anti-inflammatory or redox-modulating properties of several natural polymeric components, such as chitosan or hyaluronic acid derivatives, further enhance the therapeutic impact of the nanoparticle (Hewlings & Kalman, 2017)(Jayakumar et al., 2011). Together, antioxidant and anti-inflammatory polymeric nanoparticles help to restore redox balance, inhibit excessive inflammatory signaling, and create a favorable milieu that promotes angiogenesis, extracellular matrix (ECM) deposition, fibroblast development, and ultimately, quicker wound healing (Z. Chen et al., 2025).

4.4 Gene and siRNA-Based Polymeric Nanotherapeutics

RNA- and gene-based treatments have emerged as successful means of addressing the molecular issues underlying chronic diabetic wounds (Guo & Dipietro, 2010). Conversely, nucleases rapidly degrade naked nucleic acids, which are highly unstable and poorly absorbed by cells (Pack et al., 2005). These limitations are addressed by polymeric nanoparticles, which provide a biocompatible, protective, and adaptable delivery platform while enabling accurate spatiotemporal regulation of gene expression (Mitchell et al., 2021)(Saghazadeh et al., 2018). The fibroblasts, keratinocytes, macrophages, and endothelial cells are among the target cells that these nanosystems can transport plasmid DNA, microRNAs, or small interfering RNAs (siRNAs) (Cheng et al., 2013).

Polymeric nanoparticles are particularly useful for gene silencing techniques, such as the use of siRNA to suppress angiogenesis, matrix-degrading enzymes, or pro-inflammatory cytokines such as TNF-α and IL-1β (Castleberry et al., 2016)(Jayaraman et al., 2012). siRNA-loaded nanoparticles help reduce inflammation, restore ECM homeostasis, and promote angiogenesis by reducing the levels of these detrimental targets (Lan

et al., 2021). Similarly, therapeutic genes encoding VEGF, PDGF, or HIF-1α can accelerate tissue regeneration and encourage neovascularization (Gainza, Villullas, et al., 2015).

Targeting ligands, such as peptides, hyaluronic acid, or antibodies, enhances cell specificity, improves transfection efficiency, and reduces off-target effects (Alexis et al., 2008). Polymeric nanoparticles provide more targeted treatment by releasing nucleic acids in response to specific wound triggers (such as low pH, high ROS, or increased enzyme activity) (Liang et al., 2021). Polymeric nanotherapeutics based on genes and siRNA are generally a promising new field for regenerative and customized wound treatment (Rodrigues et al., 2019).

4.5 Cell and Exosome Integrated Polymeric Nanoparticles

A complex therapeutic strategy for diabetic wound healing is cell- and exosome-integrated polymeric nanoparticles, which combine the stability, adaptability, and controlled release properties of polymeric nanocarriers with the biological effectiveness of living cells or their secreted vesicles (Saghazadeh et al., 2018)(Gainza, Villullas, et al., 2015). The production of a variety of growth factors, cytokines, and extracellular vesicles by fibroblasts, keratinocytes, endothelial cells, and mesenchymal stem cells (MSCs) naturally encourages angiogenesis, immunomodulation, and tissue regeneration (Rodrigues et al., 2019)(Martin & Nunan, 2015). However, applying cells or exosomes directly to chronic wounds frequently fails because they are fast-acting, short-lived, and ineffective as therapies (B. C. Dash et al., 2018). Polymeric nanoparticles circumvent these issues by encapsulating themselves in exosomes or cell-derived membranes, thereby increasing their stability, bioavailability, and capacity to engage with cells at the wound site (Fang et al., 2018).

Exosome-loaded polymeric nanoparticles benefit from exosomes' inherent regenerative capabilities, which include conveying miRNA, influencing the immune system, and promoting fibroblast activity (Vader et al., 2016). Additionally, they ensure that the exosomes are released gradually and shielded from enzyme degradation (Kalluri & LeBleu, 2020). Pro-healing chemicals such as VEGF, TGF-β, miR-21, and miR-126 are particularly abundant in MSC exosomes (L. Hu et al., 2016)(Shi et al., 2017).

When these compounds are incorporated into nanoparticles, they accelerate ECM remodeling, encourage angiogenesis, and prolong their presence in the wound bed(C. Wang et al., 2019). Cell membrane-containing nanoparticles are used in another novel technique. Membranes from MSCs, macrophages, or fibroblasts wrap the polymeric cores of these nanoparticles (such as PLGA, PEG, or PCL) (L. Li et al., 2018). These biomimetic

devices aid in targeted regeneration, immune system mobility, and cell navigation to inflammatory wound areas (Thamphiwatana et al., 2017).

Polymeric nanoparticles integrated with cells and exosomes combine the engineering precision of nanotechnology with the biological power of cell-based therapy (Langer & Tirrell, 2004). Because of this, they are a very promising and clinically useful method for promoting full-thickness tissue regeneration, reducing inflammation, and restoring blood flow in diabetic wounds (J. Boateng & Catanzano, 2015). The next development in multi-functional, biologically intelligent wound-healing nanotherapeutics is these state-of-the-art hybrid systems (Yu et al., 2021).

5. Smart and Stimuli-Responsive Polymeric Nanoparticles

An innovative, cutting-edge method for treating diabetic wounds is the use of polymeric nanoparticles integrated with cells and exosomes (Saghazadeh et al., 2018)(Gainza, Villullas, et al., 2015). They combine the stability and regulated distribution of nanoscale polymeric structures with the regenerative potential of physiologically active vesicles. Growth factors, cytokines, microRNAs, and proteins that naturally promote angiogenesis, reduce inflammation, and stimulate fibroblast proliferation are abundant in exosomes derived from mesenchymal stem cells (MSCs) (Kalluri & LeBleu, 2020)(L. Hu et al., 2016). However, they are less effective as therapies due to their short half-life and rapid degradation in the wound environment (Vader et al., 2016). Exosomes become more stable, release more slowly, and remain at the wound site for longer when they are coated or encapsulated in polymeric nanoparticles such as PLGA, chitosan, or hyaluronic acid-based carriers (Yu et al., 2021)(Kim et al., 2020). Targeted engagement with inflammatory or diseased tissues is enabled by cell membrane-coated nanoparticles, such as those derived from macrophages, platelets, or stem cells, which mimic natural cellular communication and promote immunological regulation (Fang et al., 2018)(L. Chen et al., 2021). These bioinspired hybrid nanoplatforms can penetrate deeper into wounds, deliver regenerative signals more effectively, and evade immune system clearance (Thamphiwatana et al., 2017). By fusing the technical precision of polymeric nanoparticles with the biological activity of cells and exosomes, these systems significantly enhance neovascularization, re-epithelialization, and overall wound healing (Martin & Nunan, 2015). They are therefore highly helpful in the treatment of long-term diabetic wounds.

6. Advanced Polymeric Nanostructured Wound Dressings

Advanced polymeric nanostructured wound dressings are multi-functional platforms that promote quicker wound healing in people with diabetes by combining biocompatible polymers with nanoscale engineering (J.

Boateng & Catanzano, 2015). In addition to covering the wound, these nanostructured systems also deliver therapeutic drugs, regulate moisture, allow oxygen to diffuse, and directly interact with the wound microenvironment to promote healing (Moura et al., 2013)(Wu et al., 2017). In addition to increasing surface area and mimicking the native extracellular matrix (ECM), their nanoscale structure, which can take the form of nanofibers, hydrogel-nanoparticle composites, sprayable nanoformulations, or 3D-printed scaffolds, also supports cell adhesion, growth, and migration (Liang et al., 2021)(Yayehrad et al., 2023)(Xue et al., 2019).

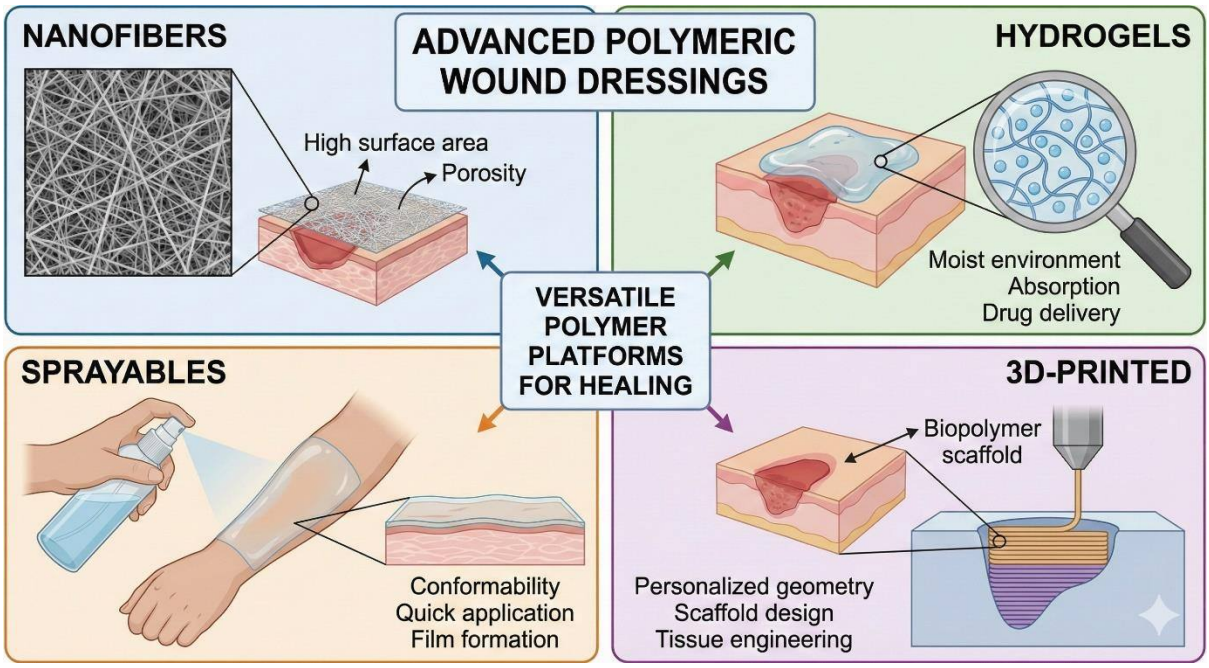


Figure 4 Diverse Formulations of Advanced Polymeric Wound Dressings

6.1 Nanofiber Mats

Nanofiber mats are among the most advanced types of polymeric wound dressings, characterized by their high surface-to-volume ratio, significant porosity, and structural similarity to the natural extracellular matrix (Xue et al., 2019)(Bhardwaj & Kundu, 2010). These factors collectively facilitate the expedited healing of diabetic lesions. Electrospinning, which enables control of fiber diameter, pore size, and surface morphology, is predominantly used to fabricate these mats (Greiner & Wendorff, 2007). Consequently, scaffolds that replicate the architecture of natural cutaneous tissue can be developed to facilitate moisture retention, nutrient distribution, and oxygen exchange (Sill & von Recum, 2008). Polymeric nanofibers made from poly(ε-caprolactone) (PCL), poly(lactic-co-glycolic acid) (PLGA), chitosan, collagen, gelatin, and polyurethane have been examined for their compatibility with living organisms, degradation properties, and mechanical strength

(Rieger et al., 2013)(K. et al., 2005). Because of their large surface area, which can retain a high amount of pharmaceuticals, nanofiber mats are excellent at delivering medications.

Additionally, they are connected by a porous network that allows chemicals such as growth factors, anti-inflammatory medicines, antibiotics, antioxidants, and even biological cargo like exosomes to be released gradually and in particular locations (J. Boateng & Catanzano, 2015)(Liang et al., 2021). To enhance antibacterial, angiogenic, and regenerative capabilities, metallic nanoparticles (e.g., Ag or ZnO), bioactive peptides, or polymeric nanoparticles loaded with growth factors can be added (Rigo et al., 2013)(Augustine, Dominic, et al., 2014). Because of this, hybrid nanofiber structures are particularly helpful for wounds caused by chronic diabetes. Moreover, nanofiber mats that respond to pH, temperature, ROS, or enzyme changes in the wound microenvironment enable intelligent, on-demand medication release (Cleeton et al., 2019)(Shafiq et al., 2021). These mats' mechanical strength provides structural support to diabetic tissue that is brittle, and their biomimetic properties facilitate fibroblast adhesion, migration, and collagen formation (Ma, 2004). Nanofiber mats can significantly accelerate wound closure, granulation tissue development, and re-epithelialization.

6.2 Hydrogel–Nanoparticle Composite Dressings

Advanced wound treatment for diabetic ulcers shows great promise with hydrogel-nanoparticle composite dressings. This is due to their ability to combine the targeted therapeutic powers of polymeric nanoparticles with the hydrating, biocompatible, and moisture-retentive qualities of hydrogels. The three-dimensional matrix produced by hydrogels, which can be derived from chitosan, alginate, hyaluronic acid, gelatin, or synthetic polymers like PEG and PVA, resembles and feels like tissue (Ahmed, 2015). They prevent pathogens from entering the wound, absorb fluids, and keep it moist. Encapsulated medications, such as growth factors, antibiotics, antioxidants, or anti-inflammatory medicines, can be released in a regulated and sustained manner at the wound site when polymeric nanoparticles are added to these hydrophilic networks (Buwalda et al., 2014)(J. Li & Mooney, 2016). The hydrogel matrix prolongs the duration of nanoparticles' presence in the body, reduces the likelihood of burst release, and increases the availability of medications in the targeted area (Kamaly et al., 2016). On-demand therapeutic delivery based on the biochemical state of the wound is also enabled by smart hydrogels containing nanoparticles that respond to stimuli (such as pH, ROS, enzymes, or temperature) (J. Hu et al., 2012). In addition to preventing bacterial proliferation and biofilm formation, these composites can promote re-epithelialization, angiogenesis, and collagen deposition. Multiple pathogenic

barriers in diabetic wounds, such as oxidative stress, microbial infection, excessive inflammation, and delayed tissue regeneration, are addressed by a hydrogel-nanoparticle composite.

6.3 Sprayable Nanoparticle Formulations

A novel, patient-friendly method for directly delivering polymeric nanoparticles to diabetic wounds with improved coverage, adhesion, and therapeutic precision is via sprayable nanoformulations (Jones et al., 2006)(Sood et al., 2014). These formulations often contain nanoparticles blended with biocompatible liquid matrices, allowing the particles to spread uniformly over uneven wound surfaces. Compared to standard dressings, this is superior (A. Gupta et al., 2019). The composition, when sprayed, forms a thin, permeable, protective nano-layer that keeps the area moist, facilitates drug penetration, and lessens trauma from the dressing (Vowden & Vowden, 2003). Antibiotics, growth factors, antioxidants, and anti-inflammatory compounds can all be found in the spray's polymer-based nanoparticles. To combat oxidative stress, infection, and poor blood flow in diabetic wounds, these nanoparticles can be delivered in a localized, controlled manner (J. Zhang et al., 2023). Certain sprayable systems use in situ gelling polymers, such as alginate, poloxamers, and chitosan, which rapidly transform from liquids to gels when they come into contact with skin or body fluids. This prevents them from slipping off and helps them remain on the skin (Ruel-Gariépy & Leroux, 2004)(M. Dash et al., 2011). Moreover, mucoadhesive or electrostatic polymer coatings enhance nanoparticle adhesion to the wound bed, prolonging treatment duration. Since sprayable nanoformulations are easy to use, cover a wide area, don't cause discomfort when changing the dressing, and can include multiple therapeutic agents, they hold promise for managing and personalizing diabetic wounds at the point of care (S. Gupta et al., 2017)(Holloway & Harding, 2022).

7. Preclinical and Clinical Evidence

Polymeric nanoparticle-based treatments for diabetic wound healing have been thoroughly tested in in vitro, in vivo, and early clinical trials, demonstrating their ability to overcome the biological barriers associated with chronic wounds (Frykberg & Banks, 2015). Preclinical research has shown that polymeric nanoparticles dramatically enhance angiogenesis, cellular proliferation, antimicrobial activity, and extracellular matrix remodeling, thereby increasing wound closure rates.

Studies conducted in vitro using endothelial cells, fibroblasts, keratinocytes, and macrophages have repeatedly

shown that polymeric nanoparticles improve drug stability, promote controlled release, and increase cellular absorption. Endothelial tube formation, fibroblast migration, and keratinocyte growth are all significantly accelerated by nanoparticles loaded with growth factors, such as PLGA or chitosan carriers containing VEGF, bFGF, or EGF (Barrientos et al., 2008)(Gainza, Villullas, et al., 2015)(Xie et al., 2013). Staphylococcus aureus, Pseudomonas aeruginosa, and MRSA are among the common diabetic wound infections that antimicrobial nanoparticles effectively kill (Rai et al., 2012)(Kalhapure et al., 2015). At the same time, antioxidant-loaded nanoparticles lower oxidative stress markers and restore redox balance in damaged cells.

Studies conducted in vivo using diabetic wound models in mice, rats, and pigs provide strong evidence for the effectiveness of polymeric nanoparticles in wound healing (Broughton et al., 2006)(Martins et al., 2008). Growth factor-containing nanoparticles stimulate neovascularization, boost collagen deposition, and accelerate wound healing more quickly than free agents (Golub et al., 2010). Antimicrobial nanoparticles disintegrate biofilms, reduce bacterial populations, and clean wound beds (G. Zhao et al., 2013). In contrast, antioxidant and anti-inflammatory nanoparticles prevent cytokine storms and facilitate an earlier transition to the proliferative phase. Advanced nanostructured dressings provide extended therapeutic release and improved retention at the wound site, including sprayable nano-formulations, hydrogel–nanoparticle composites, nanofiber mats, and 3D-printed polymeric scaffolds (C. Wang et al., 2019)(Gainza, Bonafonte, et al., 2015)(Augustine et al., 2015). Better granulation tissue development and quicker re-epithelialization are typical significant outcomes in these models (Barrientos et al., 2008).

Clinical evidence, while still developing, is very promising (Game et al., 2016). Some formulations that employ polymeric nanoparticles, such as hydrogel dressings, PLGA growth factor systems, and chitosan–silver nanocomposites, have advanced to early-phase clinical studies (Khansa et al., 2019)(Choi et al., 2008). According to these investigations, there is an improvement in wound closure rate, infection rate, moisture retention, and patient comfort (Moura et al., 2013). Because of their high biocompatibility and low toxicity, polymeric nanoparticles are a promising option for routine medical applications (Alexis et al., 2008). However, issues including scaling up nanoparticle production, obtaining regulatory approval, and demonstrating their long-term safety remain unresolved (Etheridge et al., 2013). Despite these issues, there is increasing preclinical and clinical evidence that polymeric nanoparticles have the potential to revolutionize the treatment of diabetic wounds.

Table 1: Preclinical Studies on Polymeric Nanoparticles for Diabetic Wound Healing

Study	Polymeri	Polymer	Therape	Administra	Key	Significance /	Referen

Model	c Nanoparticle Type	Composition & Features	utic Agent Loaded	tion Form	Biological Outcomes	Remarks	ce
STZ-induced Diabetic Rat Model	PLGA Nanoparticles	Biodegradable, sustained release	VEGF	Topical gel	↑ Angiogenesis, ↑ capillary density, and faster granulation	Stabilizes VEGF and prolongs angiogenic signaling	(Cheredy et al., 2015) (Hirota et al., 2010)
db/db Diabetic Mice	Chitosan Nanoparticles	Mucoadhesive, cationic, antimicrobial	EGF	Hydrogel dressing	Faster re-epithelialization, ↑ keratinocyte migration	Enhances growth factor stability in protease-rich wounds	(Ezhilarasu et al., 2020) (Gainza, Villullas, et al., 2015) (Park et al., 2017)
Alloxan-Induced Diabetic Rats	Alginate–PLGA Hybrid NPs	High water-holding capacity, biocompatible	Curcumin	Film dressing	↓ ROS, ↓ inflammation (TNF-α), ↑ collagen	Improves antioxidant delivery and ECM remodeling	(Agarwal et al., 2021)
STZ Rats	PEGylated PLGA NPs	Enhanced stability, controlled release	PDGF-BB	Injectable depot	↑ Fibroblast proliferation, ↑ ECM deposition	Mimics natural PDGF gradient over days	(Yang et al., 2020)
Ex Vivo Human Skin Wounds	Hyaluronic Acid NPs	Bioadhesive, HA receptor targeting (CD44)	bFGF	Spray formulation	↑ Cell proliferation, ↑ wound closure rate	Demonstrates translational potential in human tissue	(Ezhilarasu et al., 2020) (Z. Chen et al., 2025)
Diabetic	Chitosan–	Antimicrobial + bioadhesive	Ag^{+} ions	Topical	↓ Bacterial load, ↓	Strong dual antimicrobial	(Rai et al.,

Rabbit Ear Ulcer Model	Silver NPs	polymer		cream	biofilms, ↑ and granulation	+ wound healing synergy	2012) (Qin et al., 2022)
db/db Mice (Infected Wounds)	PLGA–Lipid Hybrid NPs	Enhanced permeability into biofilms	Antibiotics (e.g., vancomycin)	Gel	Biofilm penetration, complete bacterial clearance	Overcomes antibiotic resistance through NP targeting	(Qin et al., 2022) (Ezhilarasu et al., 2020)
STZ Rats	PCL Nanoparticles	Slow-degrading polymer, mechanical stability	Resveratrol	Nanofiber patch	↓ Oxidative stress, ↑ angiogenesis	Sustained antioxidant action is beneficial for chronic wounds	(Z. Chen et al., 2025)
Diabetic Pig Model (Large Wounds)	HA-PLGA NPs	Biomimetic ECM interaction	siRNA (TNF-α)	Hydrogel	↓ Pro-inflammatory cytokines, accelerated closure	Demonstrates targeted gene therapy feasibility	(Qin et al., 2022) (Cai et al., 2021)
3D Diabetic Wound-on-Chip	Gelatin Nanoparticles	Protease-responsive degradation	TGF-β1	Microfluidic culture	Enhanced fibroblast migration, aligned ECM	Simulates a clinical chronic wound microenvironment	(Kumar, Kumar, et al., 2024)
MRSA-Infected Diabetic Mice	Chitosan–ZnO NPs	Metal oxide antimicrobial + polymer matrix	ZnO	Dressing	Strong bactericidal action, ↓ inflammation	Useful in drug-resistant infections	(Ghazzy et al., 2023)
STZ Rats	Smart ROS-Responsive NPs	Thioketal polymer shell, ROS-triggered release	Curcumin + NO donor	Hydrogel	Reduced oxidative burden, improved vasodilation	Dual ROS scavenging + NO improves perfusion	(Joorabloo & Liu, 2024)

db/db Mice	Exosome-Coated Polymeric NPs	Enhanced targeting and immune modulation	miR-21	Topical gel	↑ Angiogenesis, ↑ fibroblast activity	Biomimetic surface enhances uptake and retention	(Ghosal et al., 2022) (W. Wang et al., 2026)
Diabetic Rats	GelMA–PLGA NPs	Photocrosslinkable hydrogel network	Antibiotic + antioxidant	Patch	↓ Infection + ↓ ROS simultaneously	Multi-functional treatment for complex wounds	(D. Dong et al., 2025) (Mo et al., 2025)
Porcine Full-Thickness Wound Model	3D-printed PCL–NP scaffold	Tailored porosity, sustained NP release	VEGF + EGF combo	Implantable scaffold	Rapid vascularization, robust epithelial layer	Mimics large chronic wounds, good translational relevance	(S. Zhang & Mikos, 2020) (Karmakar et al., 2025)

8. Challenges and Future Prospects

Even though polymeric nanoparticle-based treatments for diabetic wound healing have come a long way, a few problems still make them difficult to use in real life. One of the main problems is that diabetic wounds are very complicated and different from each other. High protease activity, poor vascularization, changing pH, and constant inflammation can all make nanoparticles less stable and less effective as treatments. Manufacturing consistency remains a major challenge because it's hard to produce more nanoparticles while maintaining their size, drug loading, and release kinetics. Also, the rules for nanomedicines aren't fully set yet, which makes it hard to be sure about safety, long-term toxicity, and the approval process. Concerns regarding polymer biodegradation, potential accumulation, and unanticipated immune responses necessitate comprehensive biocompatibility and toxicological evaluation. Moreover, most contemporary studies are confined to the preclinical stage, with few large randomized clinical trials conducted to confirm therapeutic

benefits in humans. Future developments will likely focus on smart, multi-functional nanoparticles that can respond to specific wound stimuli (ROS, enzymes, temperature), incorporate biosensing capabilities, and enable real-time monitoring of the wound's condition. Nanoparticles combined with advanced wound dressings, such as nanofiber mats, 3D-printed scaffolds, and self-healing hydrogels, have significant potential for personalized, precise wound care. Combining optimization with artificial intelligence, machine learning, and digital wound-assessment tools could further speed up the process, allowing therapies to be tailored to each patient's needs. In the end, standardized manufacturing, robust clinical trials, and regulatory harmonization will help us overcome the current translational barriers. This will make it possible for next-generation polymeric nanotherapeutics to become a common treatment for diabetic wounds.

9. Conclusion

Diabetic wound healing continues to pose a significant clinical challenge owing to ongoing inflammation, compromised angiogenesis, heightened oxidative stress, microbial infection, and ineffective extracellular matrix remodeling. In the last ten years, polymeric nanoparticles have become a game-changing platform that can overcome these many barriers by releasing drugs in a controlled way, targeting delivery, and exerting multiple therapeutic effects. They can hold a wide range of substances, such as growth factors, antimicrobials, antioxidants, anti-inflammatory compounds, nucleic acids, and exosomes. This allows them to precisely modulate the wound microenvironment and enhance the ability of diabetic tissues to heal.

Advanced polymeric nanostructured dressings, including nanofibers, hydrogel–nanoparticle composites, sprayable nano-formulations, and 3D-printed scaffolds, improve local retention, moisture balance, and biological responsiveness even more than regular wound treatments. Preclinical investigations consistently demonstrate that angiogenesis, collagen deposition, re-epithelialization, and wound healing occur more rapidly. Even though clinical translation is advancing, there are still major problems that need to be solved before it can be widely used. These issues encompass large-scale production, reproducibility, long-term safety, regulatory approval, and cost-effectiveness. Polymeric nanoparticles are a promising, highly flexible approach for treating diabetic wounds in the future. Their incorporation of innovative technologies such as intelligent stimulus-responsive systems, bioengineered tissues, wearable biosensors, and tailored medicine has considerable potential to transform chronic wound management. To turn these new nanotherapeutics into reliable, effective, and easy-to-use treatments for people with diabetic wounds, we need to continue research across disciplines and ensure they work well in the clinic.

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